



Targetless /u/ in Tokyo Japanese

Shigeto Kawahara¹, Jason A. Shaw^{1,2,3}, James Whang^{1,4}
Keio University¹, Western Sydney University², Yale University³, New York University⁴

High vowel devoicing

In Tokyo Japanese, high vowels tend to devoice between voiceless consonants:



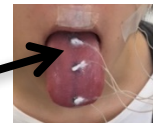
Research Questions:

Do devoiced vowels have lingual targets?

Are lingual gestures reduced when inaudible?

Electromagnetic Articulography (EMA)

Equipment: NDI Wave EMA system used to transduce movements of three fleshpoints on the tongue:



Tongue Dorsum (TD)

Participants: 6 (3 male) Tokyo natives aged 19-22

Materials: 5 near minimal pairs with voiced/voiceless /u/ read in carrier phrase: *ooke* ___ *to itte* 'okay, say ___'; 10-15 repetitions per word.

Determining target specification: DCT, Micro-prosodic sampling, Bayesian Classification

Discrete Cosine Transform (DCT) to reduce dimensionality of the data

Define stochastic generators of target and targetless hypothesis

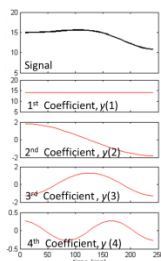
Compute Bayesian posterior probabilities of targetlessness

Complex curve represented as the sum of Cosines:

$$y(k) = w(k) \sum_{n=1}^L x(n) \cos\left(\frac{\pi(2n-1)(k-1)}{2L}\right)$$

$$k = 1, 2, \dots, L$$

Where L is the number of data samples and $x(n)$ is the trajectory to be modelled and:

$$w(k) = \begin{cases} \frac{1}{\sqrt{L}} & k = 1 \\ \frac{2}{\sqrt{L}} & 2 \leq k \leq L \end{cases}$$


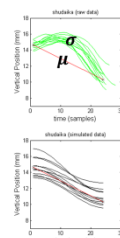
Simulate targetless trajectory from DCT coefficients:

$$y(k) \sim N(\mu(k), \sigma(k))$$

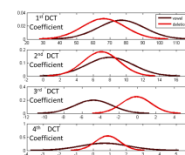
$$x(n) = \sum_{k=1}^L w(k) y(k) \cos\left(\frac{\pi(2n-1)(k-1)}{2L}\right)$$

$$n = 1, 2, \dots, L$$

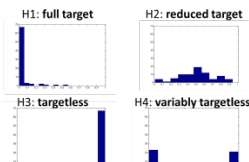
Where L is the number of data samples and $x(n)$ the trajectory to be simulated and:

$$w(k) = \begin{cases} \frac{1}{\sqrt{L}} & k = 1 \\ \frac{2}{\sqrt{L}} & 2 \leq k \leq L \end{cases}$$


Bayesian Classification parameters (for *futaise*):

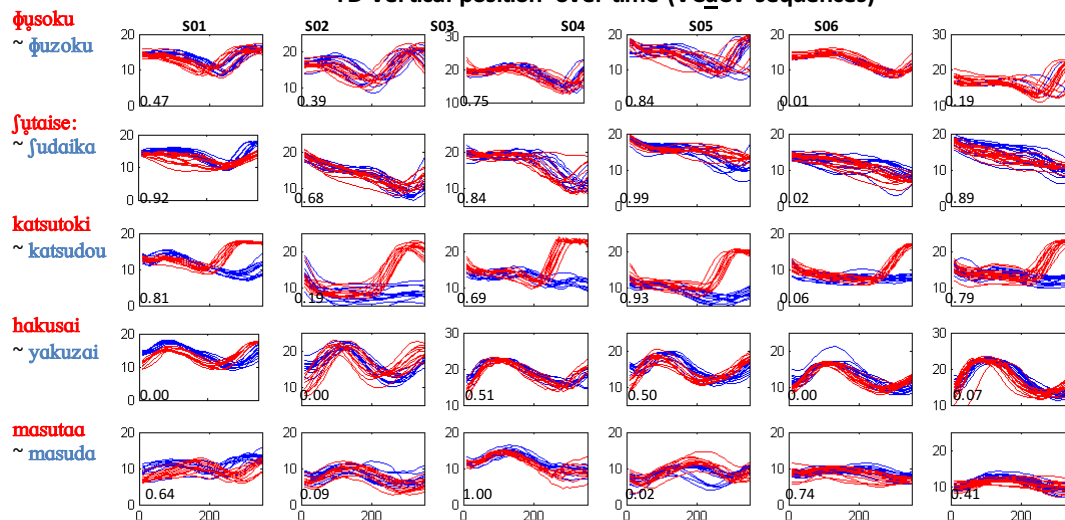


Possible outcomes expressed as posterior probabilities

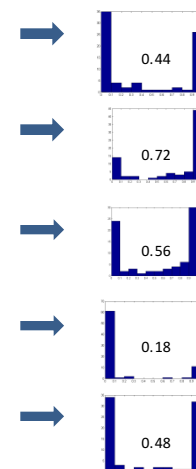


Results

TD vertical position over time (VCuCV sequences)



Posterior probability of targetlessness



Discussion

- Japanese has consonant clusters (sometimes) in that the TD follows a linear trajectory from C1 to C2 for many tokens.
- Syllable contact laws may influence the appearance of clusters: $\text{ʃ.t} > \text{ts.t} > \text{s.t}$ (exception) $> \text{ϕ.s} > \text{k.s}$
- Results support a phonological view of high vowel deletion rather than laryngeal gestural overlap (e.g., Kondo, 2001 vs. Jun & Beckman, 1993).
- The computational tools can be broadly applied to detect phonological (under-)specification in phonetic data.

Conclusions

ALL OR NOTHING

- Devoiced vowels are **variably targetless**; the probability of targetlessness:
 - varies across speakers
 - is conditioned by linguistic factors
- Phonetic reduction**, on the other hand, is uncommon.

Acknowledgments: Supported by JSPS fellowship to Jason A. Shaw (#15F15715) and grant-in-aid (#26284059) to S. Kawahara; thanks to Chika Takahashi and Jeff Moore for help with data.